

CS39440 Major Project: Rover Exploration Game/SimProject Outline

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Project Description

This project aims to develop an entertaining strategy game based upon rover exploration through feature driven development. The primary goal is develop a fun and interactive game in which the player must navigate a range of diverse environments whilst managing various resources to ensure the success of their mission, whether it be a simple scouting mission or require more complex tasks such as sample collection.

There are a number of essential aspects to this project including:

A Hexagonal Tile System: The world (levels) will be made up of a hexagonal tilemap, as the objective of the game is resource management. Allowing the rover to move in more directions allows for more creative situations and more thoughtful player responses. The rovers movement within this system will depend on various resources (yet to be determined), with each hexagon requiring either energy and/or a specific tool to navigate through.

Dynamic Terrain: The individual terrain tiles will be one of five heights, allowing for vertical navigation through within a 2D environment. To navigate to a piece of terrain the height must be either the same or adjacent. Dynamic terrain is also planned, requiring the player to react to various challenges such as hydrothermal vents or asteroids striking the planet.

Challenging Objectives: There will be a variety of missions for the player to complete such as simple navigation objectives for a tutorial to acquiring and returning samples. Finding a reasonable difficulty curve will be a challenge due to the various mechanics.

Rover Customisation (Undecided): There is a possibility for the player to design their own rover from preset parts, requiring them to think about the mission before it even begins. These could have any number of benefits or detriments, but will provide another layer to the gameplay.

The end goal of the project is to provide a fully functioning strategy game that is enjoyable to play, and retains player engagement. Various stretch goals can expand upon this, such as mobile compatibility or multiplayer.

Proposed Tasks

Feature Planning-

The task of feature planning will involve me coming up with a list and prioritising core features for the game. I've already started doing this via my git issue board, however each of these mechanics need to be researched to ensure they are possible with my tools and within the timeframe. Various other features will inevitably be thought of as the project continues, and these will have to be prioritised accordingly.

Mechanic Implementation-

Once the features have been researched and prioritised development can begin. This will start with the core features (such as movement and tile system) before moving onto the more dependent features. Stretch goals will be the last priority, as these won't massively effect the game, just provide benefits (such as mobile compatibility).

Sprites, Animation and Audio-

Sprites, animation and audio all depend on me researching online to find the ones that best fit my project. Whether they are open source or paid for will be a personal decision that occurs at the time, but in the meantime I can provide my own 2D placeholder sprites just to my familiarity with Aseprite.

Testing-

The majority of tests will be gameplay tests. Automated tests would be great to have as there will always be bugs that escape gameplay tests, but this will require research as I have never developed automated tests for use in a game. Gameplay tests will be outsourced to players of various skill levels to avoid developer bias.

Documentation-

Documentation will consist of various files. Firstly will be design files and relevant diagrams, that I can present to my supervisor before actually committing to any designs. The most important documents will be those required by the assessments such as the Project Report and Technical Work, which will be written closer to the hand in dates, based off my daily notes.

Project Deliverables

Working Rover Game- A fully functional rover strategy game with various working features that maintain player engagement. *(Produced at end of project deadline but a demo will be available for the mid demo)*

Research Documentation- Documentation on any research done into any topics such as mechanics or assets. *(Produced when topic is researched to ensure relevant credit given)*

Design Documentation- Game design documents showcasing various game aspects, such as UI design, level design and the overall visual style. *(Produced prior to the aspect being developed to ensure they are suitable before using that time)*

Testing Documentation- Documents showing feedback from tests, and an actual test table to ensure appropriate tests are performed. *(Produced as tests are written to follow TDD)*

Project Report- Will include the above documentation for research, design and testing as well as an outline of the project as a whole including how and why certain decisions were made. *(Produced near the end of the deadline based off my project notes/blogs)*

Initial Annotated Bibliography

Dagim Sileschi (2023). *Baeldung*. [online] Baeldung on Computer Science. Available at: <https://www.baeldung.com/cs/feature-driven-development>.

GeeksforGeeks. (2020). *FDD Full Form*. [online] Available at: <https://www.geeksforgeeks.org/fdd-full-form/>.

Product Plan (n.d.). *What is Feature Driven Development (FDD)? | Definition*. [online] www.productplan.com. Available at: <https://www.productplan.com/glossary/feature-driven-development/>.